World Risk Observer

Complete Professional User Guide

Step-by-Step Tutorials & Real-World Examples

Psychology Simulation

Scenario Planning

Decision Matrix

Communication

Risk Assessment

Professional Reports

Version 1.0 | 2026

World Risk Observer System

Complete Documentation with Real-World Examples

Table of Contents

1. Getting Started	Page 3
2. System Requirements & Setup	Page 4
3. Tab 1: Psychology Simulation	Page 5-6
4. Tab 2: Extract Insights	Page 7
5. Tab 3: Generate Scenarios	Page 8
6. Tab 4: Communication Strategy	Page 9
7. Tab 5: Risk Assessment	Page 10
8. Tab 6: Decision Matrix	Page 11
9. Advanced Features (Tabs 7-10)	Page 12
10. Real-World Example: Food Crisis	Page 13-14
11. FAQs & Troubleshooting	Page 15
12. Best Practices & Tips	Page 16

1 Getting Started with WRO

What is WRO?

World Risk Observer (WRO) is a comprehensive AI-powered crisis management platform that helps organizations analyze crowd psychology, generate strategic scenarios, develop communication plans, assess risks, and make informed decisions during complex crises.

Key Capabilities

- **Psychology Simulation:** AI-driven modeling of crowd behavior with autonomous agents
- **Emotional Contagion:** Track how emotions spread through populations in real-time
- **Scenario Planning:** Generate optimistic, neutral, and pessimistic outcome forecasts
- **Communication Strategy:** Stakeholder-specific messaging plans with timelines
- **Risk Assessment:** Comprehensive probability × impact analysis
- **Decision Matrix:** Prioritized action plans with ownership and KPIs
- **Memory System:** AI learning and historical context injection
- **Professional Reports:** PDF exports with charts and full analysis

Who Should Use WRO?

- Crisis Management Teams
- Government Agencies
- Corporate Risk & Compliance Departments
- Emergency Response Coordinators
- Strategic Planning Departments
- Communications & PR Teams

System Architecture

Input Crisis → AI Analysis → Scenarios → Decision Matrix

Page 2

2 System Requirements & Setup

What You Need

Hardware Requirements

Component	Minimum	Recommended
Processor	Dual-core 2GHz	Quad-core 2.5GHz+
RAM	2GB	8GB+
Storage	50MB	200MB+
Internet	1Mbps	5Mbps+ (stable)

Software Requirements

- **Browser:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- **JavaScript:** ES6+ enabled (enabled by default)
- **Storage:** localStorage enabled (5MB+ available)
- **API:** Groq API key (free from console.groq.com)

5-Minute Setup Guide

1 **Download and Open WRO**

Download the WRO.html file and open it in your web browser. No installation required - it runs 100% in your browser.

2 **Get Your Groq API Key**

Visit **console.groq.com** and create a free account. Generate a new API key from the dashboard (takes 2 minutes).

3 **Configure API in WRO**

Go to 🧠 **Psychology** tab → 🔑 **API Configuration** section. Paste your API key and select your AI model.

4 **Test Connection**

Click "🔌 **Test Connection**" to verify your setup. You should see a success message.

💡 **TIP:**

Your API key is stored only in your browser's localStorage. It is never sent to external servers or stored online.

Available AI Models

Model	Speed	Power	Best For
Llama 3.3 70B ⭐	Medium	High	RECOMMENDED - Best balance
Llama 3.1 405B	Slower	Maximum	Complex analysis, large contexts
Mixtral 8x7B	Fast	Good	Quick analysis, cost-effective

3 Tab 1: Psychology Simulation

What It Does

The Psychology Simulation tab lets you create AI-powered simulations of how crowds behave during crises. Autonomous agents interact with each other, emotions spread through contagion, and the AI analyzes collective behavior patterns.

Step-by-Step: Running Your First Simulation

1 **Describe Your Crisis Scenario**

In the **"Crisis Scenario"** field, describe your situation in detail: What happened? Who is affected? What is the current status? What are the challenges?

📋 **Example Crisis Scenario**

"A major city experiences severe food shortage crisis due to supply chain disruption. Rice prices spike 300% in 2 weeks. Supermarkets are depleted, citizens panic buying. Long queues form from 3am. Rumors spread about government storage corruption. Local authorities struggle with response."

2 **Define Social Groups (Personas)**

List all social groups affected by the crisis. Each becomes an autonomous agent with different psychology. Separate by commas. Diverse groups capture realistic behavior.

📋 **Example Personas**

Rich hoarders, Middle-class panicked, Poor desperate, Black market traders, Local leaders, Healthcare workers, Lonely elderly, Rebellious youth

3 Set Simulation Parameters

Configure how long the simulation runs and how many agents to model.

Parameter	Range	Recommended	Impact
Time Steps	2-10	4-6	Higher = longer forecast, more computation
Agent Count	3-20	6-10	Higher = richer data, more realistic
Resource Level	0-100%	Based on scenario	Lower = more fear/aggression

4 Choose Initial Crowd Mood

Calm: Low baseline emotions (peaceful events)
Cautious: Neutral awareness (normal crisis start)
Worried: Already anxious (escalating situations)
Panicked: High baseline fear (severe emergencies)

5 Click "RUN PSYCHOLOGY SIMULATION"

The system will run the simulation (4-6 minutes). Watch real-time metrics update: Fear %, Aggression %, Empathy %, Resource Level %, Stress Level.

> ⚠ **WARNING:**
>
> Simulation requires active Groq API connection. Do not close the page during processing.

3️⃣ Psychology Simulation (Continued)

Understanding the Real-Time Metrics

- 😨 **Fear Level (0-100%):** Population's collective fear and anxiety
- 💰 **Greed Level (0-100%):** Hoarding and self-interest tendency
- ⚔️ **Aggression (0-100%):** Violence and conflict likelihood
- 🤝 **Empathy (0-100%):** Mutual aid and cooperation tendency
- 🔥 **Collective Stress (0-100%):** Overall population stress level
- 📦 **Resource Level (0-100%):** Available resources in crisis

Analysis Outputs

- 🎬 **Macro Analysis:** Collective psychology and group behavior trends
- 👤 **Micro Analysis:** Individual agent thoughts and emotional states
- 👥 **Crowd Analysis:** How emotions spread and contagion patterns
- 🔮 **Forecast:** Predicted behavior in next 24-48 hours

Testing Scenarios: Event Injection

While simulation runs, you can "inject" events to test how the crowd reacts. This helps identify tipping points and critical moments.

Available Events

Event	Effect	Triggers
📦 Resource Scarcity	↑ Fear +20%, Greed +15%	Supply failure, rationing

⚔️ Security Threat	↑ Fear +25%, Aggression +20%	Attacks, riots, violence
🎤 Leadership Speech	↓ Fear -15%, ↑ Empathy +10%	Public address, reassurance
🎁 Resource Abundance	↓ Fear -20%, ↓ Aggression -15%	Aid arrives, supply restored

Simulation Controls

Control	Function	When to Use
⏭️ NEXT TIME STEP	Advance simulation by one period	Manual step-by-step analysis
⏸️ PAUSE	Pause/resume simulation	Need time to analyze state
🔄 RESET	Clear simulation, start over	Test different setup

Exporting Results

- 📥 **Export Psychology Report:** JSON file with metrics, agents, analysis, timeline
- 💾 **Download Simulation Data:** Raw data and historical snapshots
- 📸 **Save Snapshot:** Save current state for later comparison

4 Tab 2: Extract Insights

What It Does

Transform raw simulation data into actionable insights: identify emotion leaders, at-risk groups, tipping points, contagion patterns, and intervention opportunities.

Step-by-Step: Extracting Key Drivers

1 **Copy Analysis from Simulation**

From the Psychology tab, copy the Macro Analysis, Micro Analysis, and Crowd Analysis. Paste into the "Key Insights" field.

2 **Format Your Insights**

Follow this structure for best AI results:

```
1. EMOTION LEADERS (Who drives crowd sentiment?) - Youth: High aggression, influences others toward action - Rich hoarders: High greed, spreads hoarding behavior 2. AT-RISK GROUPS (Who becomes extreme first?) - Poor desperate: Fear + desperation = violence-prone - Youth: Aggression + energy = mobilization risk 3. TIPPING POINTS (When does crowd shift?) - When resource < 30%: Cooperation → Competition - When resource < 20%: Competition → Conflict 4. CONTAGION PATTERNS (How does emotion spread?) - Fear spreads: Media → Authority → General public - Path: Fear (0-12h) → Greed (12-24h) → Aggression (24-48h) 5. INTERVENTION POINTS (What prevents escalation?) - Leadership speech: -15% fear within 2h - Resource distribution: -25% greed within 6h
```

3 **Click "EXTRACT KEY DRIVERS"**

AI analyzes your input and structures findings with psychological drivers, triggers, at-risk groups, pathways, and leverage points.

Key Insights Explained

Insight Type	Definition	Use In
Emotion Leaders	Groups influencing others most	Communication targeting
At-Risk Groups	Most vulnerable to escalation	Resource allocation
Tipping Points	Thresholds for behavior shifts	Trigger monitoring
Contagion Paths	How emotions spread	Communication strategy
Intervention Points	Where small actions have big impact	Resource prioritization

> ✓
>
> **Alternative:** Click "💡 AI SUGGEST INSIGHTS" to let AI directly analyze simulation metrics without manual formatting.

5 Tab 3: Generate Scenarios

What It Does

Create three outcome forecasts (Optimistic, Neutral, Pessimistic) based on crisis context and psychology insights. Includes multi-perspective analysis from three AI agents.

Step-by-Step: Generating Scenarios

1 **Provide Crisis Context**

Paste the crisis scenario from psychology simulation. Include: situation, affected groups, current status, and psychology metrics.

2 **Select Response Timeframe**

24 Hours: Immediate response focus
72 Hours: Short-term stabilization
1 Week: Recovery narrative begins
2 Weeks: Full recovery planning

3 **Choose Focus Area**

Crowd Psychology: Emotional focus
Economic Impact: Financial consequences
Social Stability: Community cohesion
Infrastructure: Systems and services

4 **Generate Scenarios**

Click "🚀 GENERATE 3 SCENARIOS" (1-2 minutes). AI creates detailed forecasts.

Understanding the Three Scenarios

📈 Optimistic Scenario

Timeline: Quick stabilization | **Outcome:** Minimal damage, fast recovery

Assumption: Organizations respond rapidly, leaders decisive, resources available

Use For: Best-case planning, hope scenarios, motivation

⚖️ Neutral Scenario

Timeline: Normal pace recovery | **Outcome:** Manageable impact

Assumption: Standard response capability, average coordination

Use For: Most likely outcomes, primary planning

📉 Pessimistic Scenario

Timeline: Prolonged crisis | **Outcome:** Severe damage, long recovery

Assumption: Slow response, poor coordination, resource scarcity

Use For: Worst-case preparation, contingency planning

Multi-Agent Analysis

Click "🔍 Compare Scenarios (Agentic)" for three-perspective analysis:

Agent	Perspective	Question
⚔️ Antagonist (Crisis Architect)	What could go wrong?	Dangerous assumptions? Tipping points?
🛡️ Mediator (Strategic Solver)	How do we counter?	What interventions prevent escalation?
📋 Auditor (Objective Reviewer)	What's realistic?	Which scenario is most probable?

Page 7

6️⃣-7️⃣ Tabs 4-5: Communication & Risk

Tab 4: Communication Strategy

What It Does

Creates phase-based communication plans tailored to stakeholder groups. Defines key messages, channels, timing, and success metrics for each audience.

Three Steps to Generate Plan

1 **List Stakeholder Groups**

Examples: Leadership, Media, General Citizens, Local Authorities, Healthcare Workers, Business Owners, Community Leaders, Youth Groups

2 **Describe Crisis Situation**

Current status, challenges, initial response, why communication matters

3 **Choose Timeline**

Immediate (0-2h) | Short-term (2-24h) | Medium-term (1-7d) | Long-term (1-4w)

Output includes: Strategic objectives, stakeholder-specific messages, phase-based actions, channels & timing, success KPIs.

> 💡 **TIP:**
>
> After generating the plan, click "📝 Generate Message Templates" to get: press statements, social media posts, and internal staff communications ready to

customize.

Tab 5: Risk Assessment

What It Does

Identifies all potential risks, assesses probability × impact, rates severity, suggests mitigation strategies, and creates contingency plans.

Process

1 **List All Potential Risks**

Brainstorm everything that could go wrong: financial, legal, operational, reputational, social, political

2 **Select Risk Category**

Financial | Legal | Operational | Reputational | Social | Political

3 **Choose Urgency Level**

🔴 Immediate (<6h) | 🟠 Urgent (6-24h) | 🟡 High (1-7d) | 🟢 Medium (1-4w)

4 **Generate Risk Assessment**

AI analyzes each risk for probability, impact, mitigation, and contingency plans

Risk Matrix Output

Visual showing each risk plotted on probability (x-axis) vs impact (y-axis) matrix. Highest-priority risks addressed first.

Page 8

8 Tab 6: Decision Matrix

What It Does

Integrates psychology insights, scenarios, communication, and risk assessment into ONE comprehensive decision matrix with clear actions, owners, timelines, and KPIs.

Workflow Integration

Psychology + Insights ↓ Scenarios + Analysis ↓ Communication + Risk ↓ DECISION MATRIX

How to Use

1 **Complete All Previous Tabs**

Ensure you've run: Psychology Simulation → Extracted Insights → Generated Scenarios → Created Communication Plan → Assessed Risks

2 **Click "CREATE DECISION MATRIX"**

Takes 2-3 minutes. AI synthesizes all data into prioritized actions.

3 **Review Action Items**

Immediate (0-30 min) | Urgent (30 min-2h) | Short-term (2-24h)

Decision Matrix Components

📋 Top 5 Immediate Actions (0-30 minutes)

Critical first steps that set tone and prevent escalation. Examples: Leadership statement, team activation, resource check.

⚡ Top 5 Urgent Actions (30 min - 2 hours)

Stabilization moves that contain crisis. Examples: Communication implementation, coordination center, resource deployment.

📅 Top 5 Short-term Actions (2-24 hours)

Build momentum toward recovery. Examples: Full response operation, stakeholder engagement, monitoring.

📊 Supporting Information

- **Resource Requirements:** What you need for each action
- **Stakeholder Communication:** Who needs to know what, when
- **Success Metrics (KPIs):** How to measure if actions work
- **Escalation Procedures:** When to activate contingency plans
- **Contingency Triggers:** If-then rules for scenario changes

Exporting Your Matrix

- **Export Decision Matrix:** JSON file for sharing with stakeholders
- **Generate Action Checklist:** Printable checklist with checkboxes for team tracking
- **Export Full Integrated Report:** Professional PDF with all analyses

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9-10 Advanced Features

Tab 7: Memory System

WRO maintains persistent memory of all analyses, enabling AI to learn from previous crises and provide contextually-aware recommendations.

What Gets Remembered

- **Scenarios:** All crisis situations analyzed
- **Insights:** AI-generated findings and patterns
- **Agent States:** Historical psychology snapshots
- **History:** Events and decisions made

> **INFO:**
>
> **Anti-Repeat System:** WRO prevents AI from repeating outputs. If similar, it automatically regenerates with variation. This ensures fresh insights even on similar scenarios.

Tab 8: Knowledge Base

Create organizational memory by saving notes about decisions, lessons learned, timeline events, and recommendations for future reference.

Create Knowledge Entry

Field	Purpose	Example
Title	Name your entry	"Food Crisis Timeline" or "Key Decisions"
Category	Organize entries	Context | Decisions | Timeline | Analysis | Lessons | Recommendations

Content	Main information (Markdown supported)	Detailed notes and findings
Tags	Link related entries	crisis, psychology, communication, decision

Knowledge Graph

Automatically visualizes connections between entries. Entries sharing tags are linked, creating an organizational knowledge graph showing how decisions relate to lessons learned.

Tab 9: Charts & Visualizations

Real-time charts showing psychology metrics over time, agent distributions, resource depletion, and tension heatmaps.

Available Charts

Chart Type	Shows	Use For
Line Chart	Fear, Aggression, Empathy over time	Trend analysis, identifying turning points
Bar Charts	Agent-by-agent levels	Identifying highest-risk agents
Heatmap	Tension by agent type	Spotting concentration of risk

> 💡 **TIP:**
>
> Use charts in presentations and reports to communicate complex psychology data visually to non-technical audiences.

Tab 10: Export & PDF Reports

Available Exports

- **Psychology Report** - JSON with simulation data
- **Scenarios Report** - Three outcome forecasts
- **Communication Plan** - Stakeholder messages

- **Risk Assessment** - Probability × impact analysis
- **Decision Matrix** - Prioritized actions
- **Full Integrated Report** - Complete PDF with all analyses
- **Knowledge Base** - JSON/PDF/Markdown export

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1 0 Real-World Example: Food Crisis

Complete Walkthrough from Start to Finish

Scenario: Major City Food Shortage

Rice prices spike 300%, supermarkets empty, panic buying, rumors of corruption, slow government response.

Step 1: Psychology Simulation Setup

Configuration

Scenario: [detailed description]
Groups: Rich hoarders, Middle-class panicked, Poor desperate, Youth, Leaders, Healthcare
Time Steps: 4 | Agents: 8 | Resource: 35% | Mood: Cautious

Step 2: Simulation Results Analysis

Metrics

Fear: 72% | Aggression: 65% | Empathy: 38%
Resource: 12% | Trust: 40% | Tension: 68%
Emotion Leaders: Youth → Adults, Rich → Middle class
At-Risk: Poor (desperate), Youth (aggressive)
Tipping Point: When resource < 20%

Step 3: Generate Scenarios (24-hour timeframe)

- 📈 **Optimistic:** Leadership statement + aid delivery → stable in 12h
- ⚖️ **Neutral:** Standard response → controlled in 24h
- 📉 **Pessimistic:** No coordination → riots risk in 8-12h

Step 4: Communication Plan

Timeline-Based Actions

Immediate (0-2h): Leadership press statement
Short-term (2-12h): Update on aid distribution
Medium-term (12-24h): Recovery narrative
Success Metrics: Message reach, mood improvement

Step 5: Risk Assessment

- **Highest Priority:** Civil unrest (high probability, max impact)
- **Mitigation:** Enhanced security, transparent updates
- **Contingency:** Martial law procedures if escalates

Step 6: Decision Matrix

- **Immediate (0-30 min):** Activate team, prepare statement
- **Urgent (30 min-2h):** Release communication, establish aid centers
- **Short-term (2-24h):** Distribute resources, monitor sentiment
- **Success:** Civil stability maintained, 80%+ confidence restored

Step 7: Export Report

Click "📦 Export Full Integrated Report" to download comprehensive PDF for leadership distribution.

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1 1 FAQs & Troubleshooting

Common Questions

Q: Is my API key secure?

A: Yes! Your API key is stored only in your browser's localStorage. It's never sent to external servers or stored online.

Q: How long does a simulation take?

A: Psychology simulation takes 4-6 minutes. This includes AI analysis of macro, micro, crowd, and forecast analyses.

Q: Can I run multiple simulations?

A: Yes! Each simulation is separate. Use Memory system to track and compare different scenarios.

Q: What if simulation fails?

A: Check: (1) API key is valid, (2) Internet stable, (3) Groq service up, (4) All required fields filled.

Q: How do I export reports?

A: Each tab has export buttons. Choose format (JSON/PDF/Markdown) and download. Full reports compile all analyses.

Q: Can I save my work?

A: Yes! Memory and Knowledge Base auto-save. Use "Create Backup" for full system backup.

Q: What's the best approach?

A: Complete all tabs in order: Psychology → Insights → Scenarios → Communication → Risk → Matrix. Each builds on previous.

Q: How detailed should my crisis description be?

A: The more specific, the better. Include: what happened, timeline, affected groups, current status, and challenges.

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1 2 Best Practices & Tips

Tips for Success

1. Start with Detailed Scenario Description

The more specific your crisis description, the better the AI analysis. Include: what happened, timeline, affected groups, current challenges, and any initial response.

2. Create Diverse Persona List

Include rich, poor, young, old, leaders, workers, professionals, etc. Diversity captures realistic crowd behavior and different response patterns.

3. Run Multiple Simulations

Test with different parameters. Compare results to understand sensitivity and identify which factors are most critical.

4. Test Event Injections Actively

Don't just watch simulation run passively. Actively inject events to test intervention effectiveness. This reveals what works best in your specific context.

5. Complete All Tabs in Order

Psychology → Insights → Scenarios → Communication → Risk → Matrix. Each tab builds on outputs from previous ones. Skipping steps reduces quality.

6. Document Everything in Knowledge Base

Save decisions, timeline events, lessons learned. Build organizational memory that AI can reference in future crises.

7. Share Reports Widely

Export professional PDFs for leadership, stakeholders, and team members. Charts and visualizations help non-technical audiences understand analysis.

8. Iterate and Refine

Real crises often surprise us. Use new information to refine simulations, scenarios, and plans continuously.

9. Monitor Real-Time Indicators

During actual crisis, compare real-world psychology metrics (fear, aggression, empathy) against your simulation forecasts. Adjust actions if actual differs from simulated.

10. Prepare Contingency Plans

WRO creates decision matrices, but real crises evolve unexpectedly. Have contingency triggers ready to pivot your approach if situation changes dramatically.

Advanced Tips

Sensitivity Analysis

Run the same scenario with different resource levels (0%, 25%, 50%, 100%) to understand how resources affect crowd psychology and behavior.

Persona Refinement

Start with 5-7 main groups. If results seem unrealistic, add more granular personas or remove overly general ones.

Timeline Optimization

For immediate crises (0-24h), use 24-hour scenarios. For building-pressure situations, use 1-week or 2-week scenarios to see longer-term patterns.

Communication Timing

Simulation reveals optimal timing for messages. If fear peaks in hour 3, time leadership speech for hour 2-2.5 for maximum impact.

Risk Prioritization

Don't try to mitigate all risks equally. Focus on highest-probability, highest-impact items first. Use risk matrix to guide resource allocation.

Page 13

🎓 You're Ready!

You now have all the knowledge needed to master World Risk Observer

Next Steps:

- ✅ Download WRO.html
- ✅ Get Groq API key (console.groq.com)
- ✅ Run your first simulation
- ✅ Generate scenarios
- ✅ Export professional report
- ✅ Share with your team

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Thank you for using WRO!